Human-in-the-Loop for High-Risk Actions

Example: Financial Transaction Above Threshold

High-risk actions require explicit human approval. The checkpoint cannot be bypassed by the agent.

1. AGENT INITIATES REQUEST

Financial Planning Agent

Requests a payment of $125,000 to Vendor ABC

Agent Context

- Goal: Pay invoice for services
- Amount: $125,000
- Recipient: Vendor ABC
- Source: Operating Account
- Justification: Q2 Marketing Campaign

2. RISK EVALUATION

Policy & Risk Engine

Evaluates the request against policies and risk rules

Risk Assessment

- Amount > $50,000 Threshold
- New Vendor: No
- Unusual Timing: No
- Required Approval: Yes

Immutable Decision Log

Request classified as HIGH RISK
Approval required before execution
(Non-bypassable)

HIGH RISK DETECTED

3. HUMAN-IN-THE-LOOP CHECKPOINT

Independent Approval Service
(Out-of-Band Control Plane)

Receives approval request with full context via secure, independent channel.

Approval Package

- Amount: $125,000
- From: Operating Account ****1234
- To: Vendor ABC (Acct ****9876)
- Purpose: Q2 Marketing Campaign
- Requested by Agent: Financial Planning Agent
- Risk Level: HIGH
- Timestamp: 2025-05-15 10:32:47 UTC

Human Reviewer

Authorized Finance Approver

Approve | Reject | Request Info

Approval signed with human identity (MFA)

REJECTED / EXPIRED

Action Blocked

No execution. Event logged.
Agent notified of denial.

APPROVED (SIGNED)

4. EXECUTION

Execution Gateway
(Enforcement Point)

Verifies approval toquet and signer before allowing execution.

Transaction Executor

Executes the approved payment via secure financial system.

Payment System

Payment of $125,000 processed to Vendor ABC.

Immutable Audit Log

Complete record of request, risk evaluation, human approval, and execution outcome.

HIGH-RISK ACTIONS REQUIRE HUMAN APPROVAL

- Financial transactions above threshold
- Data deletion
- Privilege modifications
- External communications to customers/partners
- Changes to production infrastructure

ARCHITECTURAL GUARANTEES

- Approval service is independent and out-of-band
- Agents cannot access, modify, or forge approvals
- All approvals require human identity with MFA
- Execution gateway enforces approval on every attempt
- All events are immutable and auditable

HOW THE FLOW WORKS

1. Agent requests a financial transaction.
2. Risk engine classifies the request and determines if approval is required.
3. If high risk, the request is sent to an independent approval service where a human reviews and decides.
4. Only a signed approval token from the human can pass the execution gateway.
5. The transaction is executed and every step is logged immutably for audit and compliance.

KEY PRINCIPLE

The human-in-the-loop checkpoint is a hard architectural constraint that cannot be bypassed by the agent, even if compromised.